



Functions, components and features 6m

Functions of a database management system (DBMS)

A DBMS is a collection of software programs that process the data and manage the database. The main functions of a DBMS are to:

- define, create and maintain a database and its structures
- store, read and manipulate database data
- perform the querying tasks
- enforce constraints on data
- maintain data integrity
- control users' access to the database and ensure database security
- carry out various database administration tasks, such as backup and recovery.

A DBMS can also enable sharing databases among multiple users and provide different views of the data for different users.

In other words, the DBMS is an interface or mediator between end users and the database. The DBMS performs all the operations on the database necessary to complete the requested tasks so that the users don't need to know the underlying intricacy of the database structure and functionality.

In many cases, the DBMS is a commercial product linked to different IT vendors, for example, Microsoft SQL Server, Microsoft Access, IBM DB2, SAP SQL Anywhere and Oracle Database. There are also open-source DBMSs, such as MySQL, PostgreSQL and SQLite.

[Skip to main content](#)

Components of a database

A typical database system (and often an information system) has five major components:

- **Hardware** – This is the equipment that provides the development, maintenance and use of a database and related software applications. Depending on the organisation's requirements, hardware can include computers, routers, switches, scanners, recognition devices (such as voice or image recognition, barcode and ID readers) and output devices (such as printers and plotters).
- **Software** – This includes several types of software, such as an operating system, the DBMS's software, database applications, data modelling tools and network software.
- **Data** – This includes the operational data and the metadata and is the reason why we have the database in the first place.
- **Procedures** – Procedures include the written or electronically held instructions (such as rules and policies) that regulate the design and use of the database system. Procedures enable users and administrators to efficiently run the database system and get accurate, timely, relevant information from the system.
- **People** – Database users, managers, data and database administrators and IT personnel (such as software developers and testers).



☒ Database components

Non-Assessed

[Skip to main content](#)

Below, you will find a list of **descriptions** and a list of **categories**. Your task is to:

- **Match** the descriptions to their corresponding category based on the role they play in a database system.

Hardware	Software	Operating procedures	Data	People

⌘

Devices that enable the input of data, such as keyboard, barcode scanners or voice recognition systems

⌘

A set of rules and policies that govern how a database system is used and maintained

⌘

The operating system that supports the database management system (DBMS)

⌘

Individuals responsible for designing, maintaining and testing the database system

⌘ Users of the system

⌘

The metadata that describes the structure and organisation of the database

⌘ Tools used to model and design the structure of a database

[Skip to main content](#)



Screens, printers and plotters used to produce physical outputs from the database system



The structured and unstructured operational data stored within the database



Network software that facilitates communication between database components



Written instructions that guide users on how to retrieve specific information from the database

Submit



Next, we will explore the structure and organisation of database tables, which serve as the foundation for storing and managing data within a database system.



Thoughts on this page?

Skip to main content



